INCH-POUND

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MILITARY SPECIFICATION

LINER MATERIAL, POLYBUTADIENE BASE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of polybutadiene base liner materials.

1.2 Classification. The liner materials shall be of the following types as specified in the contract or order (see 6.2):

Type I	CCU-22/A Impulse Cartridge, CCU-22/A/A Impulse Cartridge, or CKU-7/A Rocket Catapult liner material.
Type II	CKU-5/A Rocket Catapult and CKU-5/A/A Rocket Catapult liner material.
Type III	CKU-5B/A Rocket Catapult liner material.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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SPECIFICATIONS

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MILITARY

MIL-P-23942	Polybutadiene, Linear, Carboxyl-Terminated
DOD-C-82656	Carbon Black, Thermal
MIL-R-82657	Resin, Epoxy, Trifunctional
MIL-P-82658	Polymer, Liquid, Polybutadiene, Carboxyl-Terminated
DOD-C-82660	Chromium 2-Ethyl Hexoate
DOD-C-82665	Carbon Black
DOD-T-82666	Tris-1-(2-Methyl Aziridinyl) Phosphine Oxide

(Unless otherwise indicated, copies of military specifications are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

CHEMICAL PROPULSION INFORMATION AGENCY

CPIA Publication No. 21 JANNAF Solid Propellant Mechanical Behavior Manual

(Application for copies should be addressed to the Chemical Propulsion Information Agency, Applied Physics Laboratory, Johns Hopkins University, Johns Hopkins Road, Laurel, MD 20810.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The liner material shall be polybutadiene-based compositions as specified herein.

3.2 Chemical composition. The chemical composition shall be in accordance with table I. The ingredients shall be in accordance with the referenced documents.

TABLE I. Liner ingredients and composition.

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Material	Specification	Weight Percent		
		Type I	Type II	Type III
Liquid Polymer, CTPB	MIL-P-82658	83.312	—	86.814
Linear CTPB	MIL-P-23942 Type II	—	87.84	—
Carbon Black	DOD-C-82665	—	8.86	6.2325
Carbon Black, Thermal	DOD-C-82656	10.00		
Chromium 2-Ethyl Hexoate	DOD-C-82660	0.124	—	0.0205
Epoxy Resin, Trifunctional	MIL-R-82657	6.564	—	6.933
Tris-1-(2-Methyl Aziridinyl) Phosphine Oxide	DOD-T-82666 Grade B	—	3.30	—

3.3 Modification of liner formulation.

3.3.1 Modification of Type I liner formulation. The properties of the Type I liner may be varied by changing the ratio of epoxy resin to polybutadiene. The total weight percent of epoxy resin, catalyst, and polybutadiene must remain constant at 90.0 percent.

3.3.2 Modification of Type II liner formulation. The properties of the Type II liner may be varied by changing the ratio of Tris-1-(2-Methyl Aziridinyl) Phosphine Oxide to polybutadiene. The total weight percent of Tris-1-(2-Methyl Aziridinyl) Phosphine Oxide and polybutadiene must remain constant at 91.14 percent.

3.3.3 Modification of Type III liner formulation. The properties of the Type III liner may be varied by changing the ratio of epoxy resin to polybutadiene. The total weight percent of epoxy resin, catalyst, and polybutadiene must remain constant at 93.7675 percent.

3.4 Propellant-to-liner bond strength. Propellant-to-liner bond strength testing shall be required before a new liner ingredient lot set or liner formulation modification (see 4.3.1) is used. the propellant-to-liner bond shall exhibit cohesive failure in the propellant web when tested at 77 ± 5•F.

3.5 Workmanship. The material shall be uniform and free from contamination and foreign material that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set

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forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.4.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.4.1, 4.4.2, and 4.4.3. When specified in the contract, the contractor shall prepare a report giving the results obtained for all inspections and tests performed and a certified statement that meets all requirements of this specification (see 6.2).

4.3.1 Inspection lot. An inspection lot shall consist of a single liner mix.

4.3.1.1 Liner mix. A liner mix shall consist of one homogeneous blend of ingredients (see 3.2) manufactured at one time in one mixing vessel.

4.3.1.2 Ingredient lot. The ingredient lot set shall be a set of the materials as specified in table I for each type of liner material, where each of the materials is from a single material lot that is acceptable under the quality provisions of this specification. If one or more of the material lots is changed from an established ingredient lot set this constitutes a new ingredient lot set. If the formulation (the weight percent of the individual ingredients) established with a new ingredient lot set is maintained, no additional testing (see 3.4.1) is required. The testing of 3.4.1 shall be performed if any formulation modification is required with an ingredient lot set, or if the ingredient lot set changes.

4.4 Test methods. Unless an alternative inspection method is proposed, tests shall be performed using apparatus and procedures specified herein. Alternative inspection methods may be proposed by the supplier but must be approved by the contracting activity prior to use in acceptance testing.

4.4.1 Ingredient weight. Each liner ingredient shall be weighed to determine compliance with 3.2 prior to mixing. Weights shall be measured with equipment having an accuracy of ± 1 percent of the ingredient weight calculated from the percentages specified in the applicable table (see 3.2).

4.4.2 Raw material. Each liner raw material shall be tested to determine compliance with 3.2.

4.4.3 Bond strength. The liner bond strength specimens shall be prepared and tested in accordance with CPIA Publication No. 21, section 4.7.2. A minimum of 4 samples shall be tested to determine compliance with 3.4.1.

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4.4.3.1 Liner test specimen. The propellant from the propellant mix used to cast the test specimens shall meet the propellant mechanical properties specified in the applicable propellant specification before the liner bond test is completed. If the propellant does not meet the mechanical properties specified in the applicable propellant specification, all bond testing completed with that propellant mix shall be deemed invalid.

a. Sandblast and degrease the inside of the gripping sleeve and the anvil disc using the same solvent used on the motor tubes.

b. Coat the inside of the gripping sleeve with a thin layer of liner. Coat the exposed side of the insulator disc with liner, so that, as a minimum, the insulator disc is fully covered by liner.

c. Cure the liner at 170 ± 5•F for 14 to 18 hours.

d. Assemble the test specimen as shown on figure 4.7.2-1 of CPIA Publication No. 21.

e. Cast propellant into the steel tube and cure with the propellant mix as specified in the applicable propellant specification.

5. PACKAGING

(This section is not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The liners described herein are used between the case or tube wall and the propellant of various cartridge devices and rocket motors, specifically:

Type I	Used in the CCU-22/A Impulse Cartridge, CCU-22/A/A Impulse Cartridge, and the CKU-7/A Rocket Catapult.
Type II	Used in the CKU-5/A Rocket Catapult and the CKU-5/A/A Rocket Catapult.
Type III	Used in the CKU-5B/A Rocket Catapult.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

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b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Inspection conditions, if other than as specified (see 4.2).

d. Whether Type I, Type II, or Type III material is required (see 1.2).

e. Whether the contractor is responsible for inspection (see 4.1).

f. Whether a quality conformance test report is required (see 4.3).

6.3 Data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Ref. Paragraph	DID Number	DID Title	Tailoring
4.3	DI-T-2072	Reports, Test	—

(The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.)

6.4 Manufacturing conditions. This material has been manufactured (processed, applied, and cured) successfully under the following conditions:

a. Humidity: 50 grains of water per pound of dry air, maximum
b. Temperature: 21 to 32•C (70 to 90•F)

6.5 Shelf life. The shelf life of the liner material when kept in a frozen condition in sealed containers is 6 months.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (keyword) listing.

Impulse Cartridge
Rocket Catapult

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Preparing activity

Navy - OS

(Project 1377-NE55)

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